

Parker Ferrer

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EDUCATION

University of Georgia, Morehead Honors College

Expected December 2027; GPA: 3.8/4.0

B.S. Mechanical Engineering

B.S. Electrical & Electronics Engineering

Foundation Fellowship Class of 2028: Premier UGA undergraduate academic scholarship (25/~16,000 students)

TECHNICAL SKILLS

Software: CAD (SolidWorks [*CSWA Certified*], OnShape, Creo, Inventor, AutoCAD, Fusion 360), Fusion 360 CAM, KiCAD

Tools: CNC Mill/Router/Lathe, Manual Mill/Lathe, 3D Printer, Waterjet, Laser Cutter, Saws (Table, Miter, Band)

Programming: Python, Java, C++ (Arduino IDE), MATLAB

WORK EXPERIENCE

Caterpillar

Peoria, IL

Systems Integration Intern — Autonomy and Automation Division

May 2026 – August 2026

- Engineered custom mechanical mounting brackets and power distribution enclosures for autonomous RAM truck test rigs, using rapid CAD modeling and 3D printing to eliminate fitment issues and speed up hardware integration.
- Evaluated vehicle-integrated test equipment through failure analysis and hands-on diagnostic testing, driving physical design changes that boosted system reliability and field serviceability.
- Collaborated with cross-functional autonomy engineers to streamline wiring harness layouts and enclosure mounting, improving technician safety and reducing maintenance downtime during vehicle testing.

ERMCO Power Partners

Athens, GA

Continuous Improvement Intern

May 2025 – August 2025

- Advanced quality-control automation via computer vision by analyzing and labeling >13,000 components, troubleshooting system integration, and coordinating with developers to refine algorithms and implementation.
- Designed, fabricated, and tested three iterations of passive, noninvasive alignment guides that improved transformer rotation consistency for automated vision inspection, achieving $\pm 3^\circ$ accuracy on >95% of units.
- Redesigned coil oven transport carts with V-groove wheel and rail systems, producing CAD drawings for fabrication and validating configurations that informed proposals to increase line efficiency and reduce handling variability.

AutomationDirect.com

Cumming, GA

Engineering Design Intern

June 2023 – August 2024

- Designed and tested a range of industrial automation and control demonstrations to showcase products.
- Spearheaded the mechanical development of a novel large-format ($>4\text{ m}^3$) 3D printer by designing the full superstructure, stepper-driven cartesian motion system, segmented heated build plate, and electrical enclosure.
- Engineered the printer's custom electrical system: authored wiring diagrams, implemented 24V/48V/220V power distribution, and integrated stepper drivers, heaters, safety relays, and sensor/controls communication.

PROJECTS & LEADERSHIP

PedalSwap

Athens, GA

Founder & Lead Product Design Engineer

November 2025 – Present

- Architected the electromechanical layout for a modular guitar pedal platform, executing 12+ rapid CAD and 3D-printing iterations in one week to convert a breadboard circuit into a functional prototype that placed 2nd place at Georgia Tech's Idea to Prototype competition.
- Engineered a toolless 3D-printed magnetic connection mechanism for swappable effect modules, optimizing user ergonomics to win 2nd place, \$10,000 in cash, and patent funding at the Georgia Tech InVenture Prize.
- Accepted into Georgia Tech's CREATE-X Startup Launch accelerator to bring the product to market, driving physical design optimizations, utility patent drawing creation, and scalable manufacturing preparation.

UGA Robotics Club

Athens, GA

Multi-Project Mechanical Team Lead & Project Manager, Club Treasurer

June 2024 – Present

- Directed a 15-member interdisciplinary team through CAD design, machining, and full assembly of an autonomous robot, advancing UGA's competition standing from 30th place to 8th out of 45 teams at the IEEE Hardware Competition.
- Intentionally increased mechanical mechanism precision and hardware sensor integration, simplifying autonomous software control requirements and overcoming software team skill bottlenecks.
- Established strict cross-functional project tracking protocols and design review gates, eliminating communication breakdowns and ensuring on-time delivery of physical sub-assemblies for veteran and rookie students alike.

UGA Cultivate Lab | Undergraduate Researcher

Athens, GA

Adaptive Instrumentation Hardware Development

September 2024 – August 2025

- Rapidly prototyped a robotic guitar, delivering 4 iterations and constructing custom sub-assemblies to achieve consistent servo-driven strumming within a compressed 3-week timeline.
- Published & presented at ASEE 2025 Annual Conference "*DeSimone, W. P., & Ferrer, P. G., & Bork, S. J. (2025, June), Bridging the Gap: Autoethnographic Insights into Project-Based Learning in Electrical Engineering*"